

MINOR PROJECT REPORT

The Hospital Management System

A

Minor Project

*Submitted in partial fulfillment of the requirement for the award of
Degree of*

**BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING**



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SRI SATYA SAI UNIVERSITY OF TECHNOLOGY AND MEDICAL SCIENCES

[Established Under Act. 06 of 2014 by Govt. of Madhya Pradesh]

Approved by Madhya Pradesh Private University Regulatory Commission

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CERTIFICATE

This is to certify that the work embodies in this Project entitled **“TITLE OF THE PROJECT”** being submitted by----- in partial fulfillment of the requirement for the award of **“Bachelor of Engineering (Computer Science and Engineering)”** to **Sri Satya Sai University of Technology & Medical Sciences, Sehore(M.P.)** during the academic year 2024-2025 is a record of bonafide piece of work, carried out by him under our/my supervision and guidance in the **“Department of Computer Science and Engineering, School of Engineering, SSSUTMS, Sehore.**

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Head of Department

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DECLARATION

(Prince) , a student of Bachelor Of Engineering in Computer Science and Engineering, VI Sem, in session 2024-25 , **School of Engineering, Sehore**, hereby declare that the work presented in this Project entitled **“The Hospital Management System”** is the outcome of my own work, is bonafide and correct to the best of my knowledge and this work has been carried out taking care of Engineering Ethics. The work presented does not infringe any patented work and has not been submitted to any other university or anywhere else for the award of any degree or any professional diploma.

Date: 13/06/2026

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ABSTRACT

The Hospital Management System is a desktop-based application developed using Python, Tkinter, and MySQL. The main purpose of this project is to simplify and automate the process of managing hospital records digitally. The system helps in storing patient details, medicine information, and prescription records efficiently.

The application provides a user-friendly graphical interface where hospital staff can easily add, update, delete, and view patient records. It also generates prescriptions and stores all information securely in a MySQL database. The system reduces manual paperwork, saves time, and improves the accuracy of record management.

The project is developed using Python programming language with Tkinter for GUI design and MySQL for database connectivity. Various functionalities such as CRUD operations (Create, Read, Update, Delete), data fetching, and prescription generation are implemented in the project.

This project demonstrates the practical implementation of database management, GUI development, and object-oriented programming concepts. The Hospital Management System can be further enhanced by adding features such as login authentication, online database support, billing systems, and appointment management.

CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's modern world, hospitals and healthcare organizations require efficient systems to manage patient information and medical records. Traditional paper-based record systems are timeconsuming, difficult to maintain, and prone to errors. To overcome these problems, computerized hospital management systems are used.

The Hospital Management System is designed to manage hospital activities digitally. It helps hospital staff maintain patient records, medicine details, and prescriptions in an organized manner. The system provides quick access to patient information and improves hospital management efficiency.

This project is developed using Python, Tkinter, and MySQL. Python is used as the programming language, Tkinter is used for designing the graphical user interface, and MySQL is used for storing data in the database.

The application allows users to:

- Add patient records

- Store medicine details

- Generate prescriptions

- Update existing records

- Delete records

- Display patient data in table format

The system is easy to use and reduces the workload of hospital staff by automating manual tasks.

1.2 Purpose of the Project

The main purpose of the Hospital Management System is to digitize hospital record management and improve efficiency in handling patient information.

The project aims to:

Reduce paperwork

Store records securely

Improve data management

Save time

Provide quick access to patient information

1.3 Scope of the Project

The scope of this project includes:

Patient record management

Prescription management

Medicine information storage

Database management using MySQL

GUI-based hospital application

The project can be used in small clinics, hospitals, and medical centers for maintaining records digitally.

1.4 Problem Statement

Managing hospital records manually creates several problems such as:

Data loss

Time consumption

Difficulty in searching records

Human errors

Large paperwork

The Hospital Management System solves these issues by maintaining records digitally in a database.

1.5 Objectives of the Project

The objectives of the project are:

To create a user-friendly hospital management application.

To store patient and prescription data digitally.

To implement CRUD operations.

To improve record management efficiency.

To reduce manual errors and paperwork.

1.6 Technologies Used

Technology

Description

Python

Main programming language

Tkinter

GUI development

MySQL

Database management

mysql.connector

Database connectivity

1.7 Advantages of the System

Easy to use

Fast data processing

Secure data storage

Reduces paperwork

Saves time

Better data management

1.8 Conclusion

The Hospital Management System is an efficient software application that helps manage hospital records digitally. The system simplifies patient record management and improves the overall

working process of hospitals. The project also provides practical knowledge of Python programming, database management, and GUI development.

CHAPTER 2 : HARDWARE / SOFTWARE REQUIREMENT AND TECHNOLOGY USED

2.1 Introduction

For developing the Hospital Management System, both hardware and software components are required. Hardware requirements ensure that the system runs smoothly, while software requirements provide the necessary tools and technologies for application development.

This chapter explains the hardware requirements, software requirements, and technologies used in the project.

2.2 Hardware Requirements

The following hardware components are required for the successful execution of the Hospital Management System.

Hardware Component

Requirement

Processor

Intel Core i3 or above

RAM

Minimum 4 GB

Hard Disk

500 MB free space

Monitor

Standard Display Monitor

Keyboard

Standard Keyboard

Mouse

Optical Mouse

2.3 Software Requirements

The following software is required for developing and running the project.

Software

Purpose

Windows Operating System

Platform for execution

Python 3.x

Programming language

MySQL Server

Database management

VS Code / PyCharm

Code editor / IDE

Tkinter Library

GUI development

mysql.connector

Database connectivity

2.4 Technologies Used

The project uses different technologies for GUI design, programming, and database management.

2.4.1 Python

Python is a high-level, interpreted programming language used for developing the Hospital Management System. It is easy to learn, simple to write, and supports object-oriented programming concepts.

Features of Python

Easy syntax

Platform independent

Object-oriented

Large library support

Fast development

Advantages of Python

Easy to understand

Reduces coding complexity

Supports database connectivity

Useful for GUI applications

2.4.2 Tkinter

Tkinter is a standard Python library used for creating graphical user interfaces (GUI). It helps in designing windows, buttons, labels, text boxes, and tables.

Components Used in Tkinter

Label

Entry

Button

Frame

Text

Treeview

Scrollbar

Combobox

Advantages of Tkinter

Simple GUI development

Easy integration with Python

Lightweight framework

Interactive user interface

2.4.3 MySQL

MySQL is a relational database management system used to store hospital records permanently.

Functions of MySQL in the Project

Store patient records

Store prescription details

Update records

Delete records

Retrieve saved information

Advantages of MySQL

Secure database system

Fast data processing

Easy data management

Supports SQL queries

2.4.4 mysql.connector mysql.connector is a Python module used to connect Python applications with MySQL databases.

Purpose

Establish database connection

Execute SQL queries

Insert data

Fetch data

Update and delete records

Example Connection Code

```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="mydata"
)
```

2.5 Development Environment

The following development tools are used in the project.

Tool

Purpose

VS Code

Writing and editing code

Python Interpreter

Running Python programs

MySQL Workbench

Managing database

Command Prompt

Running scripts

2.6 Database Requirements

Database Name

mydata

Table Name

hospital

Main Fields Used

NameofTablets

ref

Dose

NumberofTablets

Lot

Issuedate

ExpDate

DailyDose

sideEffect

PatientName

PatientAddress

2.7 System Architecture

The Hospital Management System works in three layers:

User Interface LayerDeveloped using Tkinter

Allows user interaction

Application LayerDeveloped using Python

Handles logic and operations

Database LayerDeveloped using MySQL

Stores and manages records

2.8 Advantages of Used Technologies

Technology

Advantages

Python

Easy coding and fast development

Tkinter

Simple GUI creation

MySQL

Secure and reliable database

mysql.connector

Easy database integration

2.9 Conclusion

The Hospital Management System is developed using modern technologies such as Python, Tkinter, and MySQL. These technologies provide a simple, efficient, and reliable environment for managing hospital records digitally. The combination of GUI development and database management makes the application user-friendly and effective for hospital management operations.

CHAPTER 3 : WORKING AND BLOCK DIAGRAM / FLOW CHART

3.1 Introduction

The Hospital Management System is designed to manage patient and prescription records digitally. The system provides a graphical user interface where users can enter patient details, generate prescriptions, store records in a database, update existing records, and delete unnecessary records.

This chapter explains the working process of the system along with the block diagram and flow chart.

3.2 Working of the Hospital Management System

The Hospital Management System works in a simple and systematic manner. The application is connected with a MySQL database where all records are stored permanently.

The working process of the project is explained below.

3.3 Step-by-Step Working Process

Step 1: Start the Application

When the program is executed, the main window of the Hospital Management System appears on the screen.

The GUI contains:

Patient Information Section

Prescription Section

Buttons

Record Table

Step 2: Enter Patient Details

The user enters the following details:

Tablet Name

Reference Number

Dose

Number of Tablets

Patient ID

NHS Number

Patient Name

Address

Date of Birth

All information is entered through text fields and combo boxes.

Step 3: Generate Prescription

After entering details, the user clicks the Prescription button.

The system generates a prescription automatically and displays it in the prescription text area.

Step 4: Store Data into Database

When the user clicks the Prescription Data button:

The application connects with MySQL database.

SQL INSERT query is executed.

Patient records are stored in the database table.

Step 5: Display Records

The saved records are displayed in the Treeview table.

The system fetches all records from MySQL database and shows them on the screen.

Step 6: Update Records

The user selects a record from the table and modifies the required information.

After clicking the Update button:

SQL UPDATE query is executed.

Existing data is updated successfully.

Step 7: Delete Records

When the user selects a record and clicks the Delete button:

SQL DELETE query is executed.

Selected record is removed from the database.

Step 8: Clear Data

The Clear button removes all entered information from the fields and resets the form.

Step 9: Exit Application

The Exit button closes the application after confirmation from the user.

3.4 Block Diagram of Hospital Management System

Block Diagram Explanation

The block diagram represents the overall structure and working flow of the system.

Main Blocks

User Interface

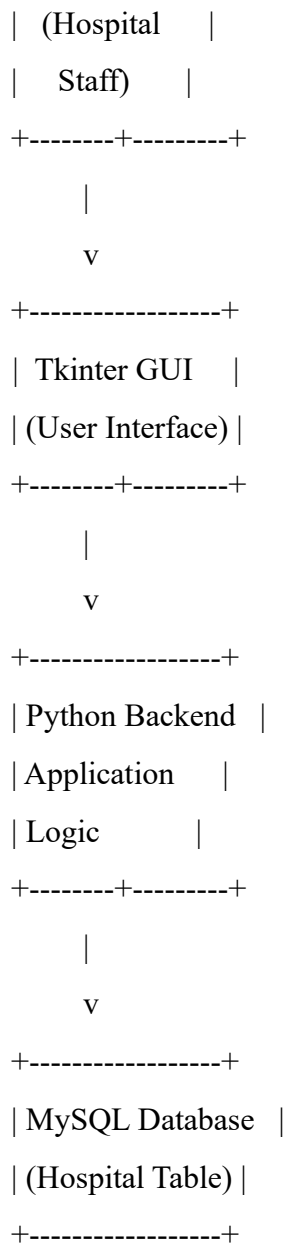
Python Application Logic

MySQL Database

Block Diagram

+-----+

| User |



3.5 Flow Chart of the System

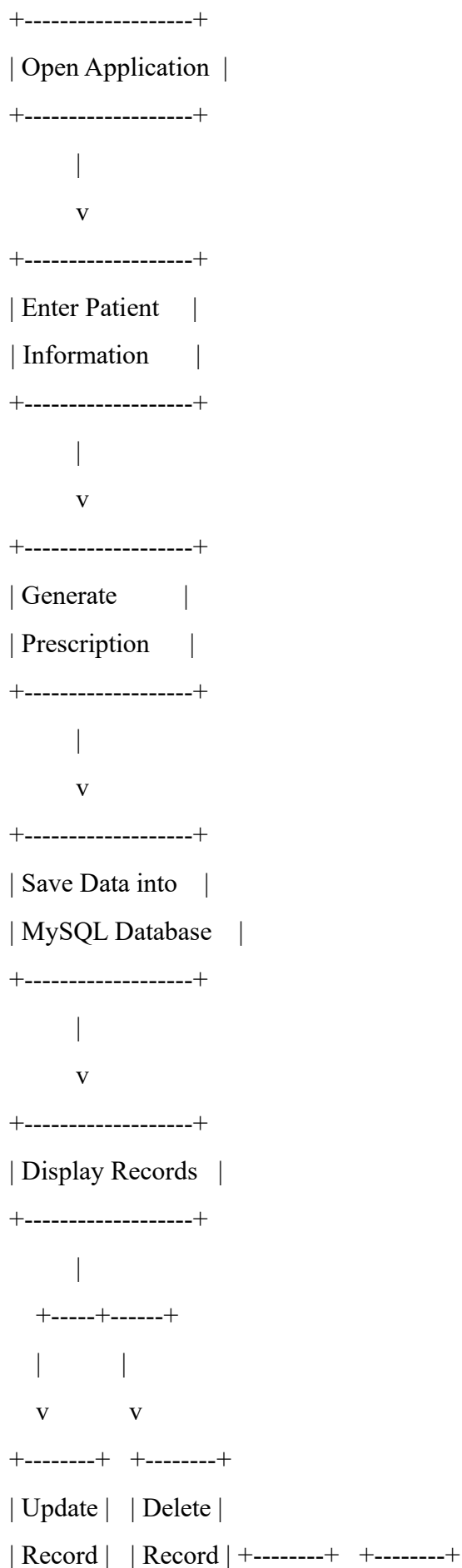
Flow Chart Explanation

The flow chart shows the sequence of operations performed in the system.

Flow Chart

START





```

      |      |
      +-----+-----+
      |
      v
+-----+
| Clear Form Data |
+-----+
      |
      v
+-----+
| Exit Application |
+-----+
      |
      v
      END

```

3.6 Functions Used in the Project

Function Name

Purpose

iPrescriptionData() Insert

records into database

update_data() Update

records fatch_data() Fetch

records from database

get_cursor() Select row data

from table idelete() Delete

records clear()

Clear form fields

iPrectioption()

Generate prescription

iExit()

Exit application

3.7 CRUD Operations in the Project

The project performs all CRUD operations.

Operation

Description

Create

Insert new records

Read

Display records

Update

Modify existing records

Delete

Remove records

3.8 Advantages of System Working

Fast processing

Easy record management

Less manual work

Better data organization

Quick data retrieval

Improved efficiency

3.9 Conclusion

The Hospital Management System works efficiently by integrating Python GUI with MySQL database connectivity. The system automates hospital record management and provides an easy way to store, update, retrieve, and delete patient information. The block diagram and flow chart help in understanding the overall workflow and structure of the project clearly.

CHAPTER 4 : RESULT ANALYSIS (SNAPSHOTS OF IMPLEMENTATION WITH DETAILS)

4.1 Introduction

The Hospital Management System was successfully developed using Python, Tkinter, and MySQL. The application performs all major hospital record management operations such as adding patient details, generating prescriptions, updating records, deleting records, and displaying stored information.

This chapter explains the implementation results and different snapshots of the system interface.

4.2 Main Window of the Application

The main window of the Hospital Management System appears after running the Python program.

The window contains:

Title section

Patient Information section

Prescription section

Buttons section

Database records table

Features of Main Window

User-friendly interface

Organized layout

Easy navigation

Fast data entry

Snapshot 1: Main Interface

```
-----
|      HOSPITAL MANAGEMENT SYSTEM      |
|-----|
Patient Information | Prescription Section |
|               |               |
| Name of Tablets |               |
| Reference No    |               |
| Dose           |               |
| Patient Name    |               |
| NHS Number     |               |
| Address        |               | | | | |
|---|---|---|---|---|---|
| Prescription | Save | Update | Delete | Clear | Exit |
|-----|
|      Records Table Section      |
|-----|
```

4.3 Patient Information Entry

The system allows users to enter patient and medicine information through input fields.

Information Entered

Tablet Name

Dose

Lot Number

Expiry Date

Patient Name

Patient Address

NHS Number

Result

The system accepts user input successfully and stores it temporarily before saving into the database.

Snapshot 2: Data Entry Form

Patient Name : Rahul Sharma
Patient ID : P101
NHS Number : NHS2025
Tablet Name : Acetaminophen
Dose : 2 Tablets
Address : Bhopal, MP

4.4 Prescription Generation

When the user clicks the Prescription button, the prescription details are generated automatically in the text area.

Generated Information

Medicine Name

Dose

Daily Dose

Patient Information

Result

The prescription generation function works successfully.

Snapshot 3: Prescription Output

Name of Tablets : Acetaminophen
Reference No : REF101
Dose : 2 Tablets
Daily Dose : Morning and Night
Patient Name : Rahul Sharma

NHS Number : NHS2025

4.5 Database Storage Result

The project successfully stores records into the MySQL database using SQL INSERT queries.

Result

Data inserted successfully

Records stored permanently

Database connection established properly

SQL Operation Used

INSERT INTO hospital VALUES(...);

4.6 Display of Records

The application displays all stored records in the Treeview table.

Features

Scrollable table

Multiple record display

Row selection support

Result

The fetch operation successfully retrieves all records from the database.

Snapshot 4: Records Table

Ref No	Patient Name	NHS Number	Dose	Tablets	

REF101	Rahul Sharma	NHS2025	2	10	
REF102	Aman Verma	NHS2026	1	5	

4.7 Update Record Result

The update feature allows modification of existing records.

Working

User selects a record

Changes data

Clicks Update button

Result

The selected database record is updated successfully.

SQL Query Used

```
UPDATE hospital SET PatientName='Rahul'
```

```
WHERE ref='REF101';
```

4.8 Delete Record Result

The delete feature removes selected patient records from the database.

Working

User selects a row

Clicks Delete button

Result

The selected record is removed successfully.

SQL Query Used

```
DELETE FROM hospital WHERE ref='REF101';
```

4.9 Clear Button Result

The clear button removes all text field values and resets the form.

Result

Form cleared successfully

Prescription text area cleared

Ready for new data entry

4.10 Exit Function Result

The exit button closes the application after confirmation.

Result

Application closes safely

Prevents accidental closing

4.11 Testing Results

Function

Status

Insert Data

Successful

Fetch Data

Successful

Update Data

Successful

Delete Data

Successful

Prescription Generation

Successful

Clear Form

Successful

Database Connectivity

Successful

4.12 Advantages Observed During Implementation

Fast response time

Accurate data storage

Simple GUI interaction

Efficient database operations

Easy record management

4.13 Limitations Observed

No login authentication

No online database support

Single-user access only

Basic GUI design

4.14 Conclusion

The Hospital Management System was implemented successfully and all major functionalities worked properly. The application efficiently manages patient and prescription records using Python and MySQL. The results obtained during implementation show that the project is reliable, user-friendly, and useful for digital hospital record management.

CHAPTER 5 : CONCLUSION AND FUTURE SCOPE OF PROJECT

5.1 Conclusion

The Hospital Management System is a desktop-based application developed using Python, Tkinter, and MySQL. The project was designed to simplify hospital record management and reduce manual paperwork. The system successfully manages patient information, medicine details, prescriptions, and database records efficiently.

The application provides an easy-to-use graphical user interface that allows users to:

- Add patient records

- Generate prescriptions

- Store records in a database

- Update patient information

- Delete unnecessary records

- Display saved records

The integration of Python with MySQL enables secure and reliable storage of hospital data.

Tkinter helps in creating an interactive and user-friendly interface for smooth operation of the system.

The project demonstrates practical implementation of:

- Python programming

- GUI development

- Database connectivity

- CRUD operations

- Object-Oriented Programming concepts

The Hospital Management System improves efficiency, saves time, reduces human errors, and provides better management of hospital records. Therefore, the project fulfills its objectives successfully and can be useful for small hospitals, clinics, and healthcare centers.

5.2 Achievements of the Project

The following objectives were successfully achieved in the project:

Development of a GUI-based hospital application

Successful MySQL database connectivity

Implementation of CRUD operations

Prescription generation system

Record display using Treeview table

Efficient patient data management

5.3 Future Scope of the Project

The current project is a basic hospital management application. In the future, several advanced features can be added to improve the system further.

5.3.1 Login Authentication System

A secure login system can be added for:

Admin

Doctors

Reception staff

This will improve system security and data privacy.

5.3.2 Online Database Integration

The project can be connected with a cloud database for online access and real-time data synchronization.

5.3.3 Appointment Management System

An appointment booking feature can be added for managing:

Doctor schedules
Patient appointments
Appointment timings

5.3.4 Billing System

A billing module can be included to:

Generate bills
Calculate medicine costs
Manage payment records

5.3.5 Doctor Management Module

Future versions can include:

Doctor information
Department details
Doctor availability

5.3.6 Report Generation

The system can generate:

Patient reports
Prescription reports
Monthly hospital reports
PDF reports

5.3.7 Multi-user Support

Currently, the system supports only one user at a time. Multi-user functionality can be implemented for larger hospitals.

5.3.8 Better GUI Design

Advanced GUI frameworks can be used to improve the user interface and user experience.

Examples:

- * Custom themes
- * Modern dashboard
- * Attractive layouts

5.3.9 Data Backup and Recovery

Automatic backup and recovery systems can be added to protect important hospital data.

5.3.10 SMS and Email Notification System

Future versions can send:

- * Appointment reminders
- * Prescription notifications
- * Emergency alerts

5.4 Final Summary

The Hospital Management System is an efficient and reliable software solution for managing hospital records digitally. The project successfully combines Python programming, GUI development, and database management into a single application.

The system reduces manual work and provides faster, more accurate, and organized record management. With future enhancements, the project can be transformed into a complete hospital automation system suitable for real-world healthcare environments.

CHAPTER 6 : REFERENCES

The following books, websites, and online resources were referred to during the development of the Hospital Management System project.

6.1 Books References

1. Python Programming

Book Name: Python Programming: An Introduction to Computer Science

Author: John Zelle

Purpose: Used for learning Python programming concepts and syntax.

2. Database Management System

Book Name: Database System Concepts

Author: Abraham Silberschatz, Henry F. Korth, S. Sudarshan

Purpose: Used for understanding database concepts and SQL operations.

3. Object-Oriented Programming

Book Name: Programming in Python 3

Author: Mark Summerfield

Purpose: Used for learning object-oriented programming concepts in Python.

6.2 Online References

1. Python Official Documentation

Python Official Website

Purpose:

Python syntax

Libraries

Programming concepts

2. Tkinter Documentation

Tkinter Documentation

Purpose:

GUI design

Tkinter widgets

Event handling

3. MySQL Official Website

MySQL Official Website

Purpose:

Database creation

SQL queries

Database management

4. MySQL Connector DocumentationMySQL Connector/Python Documentation Purpose:

Python and MySQL connectivity

Database operations

5. W3Schools Python Tutorial

W3Schools Python Tutorial

Purpose:

Python basics

GUI examples

SQL tutorials

6. GeeksforGeeks

GeeksforGeeks

Purpose:

Python programming examples

Tkinter examples

Database connectivity tutorials

6.3 Software Tools Used

Software Tool

Purpose

Python 3.x

Programming language

Tkinter

GUI development

MySQL

Database management

VS Code

Code editor

MySQL Workbench

Database management tool

6.4 Conclusion

The references mentioned above helped in understanding Python programming, GUI development using Tkinter, and database management using MySQL. These resources were useful during the design, development, implementation, and testing phases of the Hospital Management System project.

APPENDIX

Appendix A : Source Code Snippets

The following code snippets were used in the development of the Hospital Management System project.

A.1 Import Libraries

The required Python libraries are imported for GUI development and database connectivity.

```
from tkinter import * from
tkinter import ttk from tkinter
import messagebox import
mysql.connector import random
import time import datetime
```

A.2 Database Connection Code

This code connects the Python application with the MySQL database.

```
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="your_password",
    database="mydata"
)
```

A.3 Insert Data Query

This query inserts patient records into the database.

```
query = ""  
INSERT INTO hospital(  
    Nameoftablets,  
    ref,  
    Dose,  
    NumberofTablets  
)  
VALUES(%s,%s,%s,%s)  
""
```

A.4 Update Data Query

This query updates existing patient records.

```
UPDATE hospital SET  
PatientName=%s  
WHERE ref=%s
```

A.5 Delete Data Query

This query deletes records from the database.

```
DELETE FROM hospital WHERE ref=%s
```

A.6 Fetch Data Query

This query retrieves all records from the database.

```
SELECT * FROM hospital
```

Appendix B : Output Screens

The following outputs were obtained during project implementation.

B.1 Main Interface

Features:

Patient Information Form

Prescription Area

Buttons

Record Table

B.2 Prescription Output

Displays:

Medicine Name

Dose

Patient Information

Prescription Details

B.3 Database Table Output

Shows:

Stored patient records

Medicine details

NHS Number

Address information

Appendix C : Sample Test Data

Field Name

Sample Value

Patient Name

Rahul Sharma

Patient ID

P101

NHS Number

NHS2025

Tablet Name

Acetaminophen

Dose

2 Tablets

Address

Bhopal, MP

Appendix D : Advantages of the Project

Digital record management

Easy data handling

Reduced paperwork

Fast processing

User-friendly interface

Appendix E : Limitations of the Project

No login authentication

Local database only

Basic GUI design

Single-user support

Appendix F : Future Improvements

Future enhancements may include:

Online database support

Appointment booking system

Billing management

Doctor management module

Report generation system

Multi-user access

Appendix G : Abbreviations Used

Abbreviation

Full Form

GUI

Graphical User Interface

DBMS

Database Management System

SQL

Structured Query Language

OOP

Object-Oriented Programming

NHS

National Health Service

Appendix H : Conclusion

The appendix section contains additional information related to the Hospital Management System project, including source code snippets, sample outputs, test data, abbreviations, and future enhancements. These details provide better understanding of the implementation and functionality of the project.